

matches the `firmware_information` field in `attestation-elements`.

4. If verification fails, report error to the user
5. If verification succeeds, proceed with device commissioning

Below is an example of the corresponding Device actions. For illustrative purposes, it is assumed that the device implements a secure subsystem that maintains the private device attestation key and signs `attestation-elements` using this key but it does not have direct hardware support for measure boot. This is expected to be the common case for many devices covered by this version of the specification. Consequently, the measurement process can only start from the bootloader shown in the figure above.

1. The device bootloader produces a measurement of the OS kernel using a supported hash algorithm from RFC 5912 and delivers it to the secure subsystem.
2. The secure subsystem receives the measurement and stores in a location inaccessible to the OS.
3. The OS kernel produces a hash of the root filesystem and delivers the measurement to the secure subsystem.
4. When the secure subsystem is asked to sign an `attestation-elements` structure using its private device attestation key, it generates two `FirmwareDigests` or one combined `FirmwareDigest` from these measurements, fills the `firmware_information` field in `attestation-elements` using these measurements, fills the CD blob compiled into the secure environment and signs the `attestation-elements` structure.

The Device Vendor is responsible to provide the `FirmwareInformation` field when a new Software Image entry is reported in the corresponding `Distributed Compliance Ledger` entry.

Below is an exemplary ASN.1 schema for an encoding scheme that could be used to encode firmware information.

Firmware Information encoding example

```

HashAlgorithm ::= SEQUENCE {
    id OBJECT IDENTIFIER,
    params ANY OPTIONAL
}

FirmwareDigest ::= SEQUENCE {
    digestAlgorithm HashAlgorithm,
    digestHash      OCTET STRING
}

FirmwareInformation ::= SEQUENCE {
    firmwareDigests SEQUENCE OF FirmwareDigest
}

-- Example HashAlgorithm id

```